

DR. REBECCA A. BOLINGER

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RESEARCH & SERVICE INTERESTS

Climate variability and extremes, drought, hydroclimate of western U.S., climate change science and impacts, climate communication

EDUCATION

Ph.D. Atmospheric Science, Colorado State University, Fort Collins, CO (2014)

M.S. Meteorology, Florida State University, Tallahassee, FL (2007)

B.S. Meteorology, Metropolitan State University of Denver, Denver, CO (2002)

POSITIONS

- [2024 – 2026] Senior Climate and Water Resources Scientist, Lynker
- [2023 – 2024] Research Scientist III, Assistant State Climatologist, Department of Atmospheric Science, Colorado State University
- [2018 – 2023] Research Scientist II, Assistant State Climatologist, Department of Atmospheric Science, Colorado State University
- [2016 – 2018] Research Associate III, Department of Atmospheric Science, Colorado State University
- [2016] Project Scientist I, University Corporation for Atmospheric Research
- [2014 – 2016] Postdoctoral Fellow, Great Lakes Environmental Research Laboratory, U.S. Army Corps of Engineers – Detroit
- [2007 – 2009] Assistant Research Climatologist, Illinois State Water Survey, University of Illinois

SKILLS AND EXPERIENCE

- 16 years of experience monitoring drought, contributing to the U.S. Drought Monitor process, and developing drought and water resource tools and products.
- 20 years of experience researching and analyzing climate variability and climate extremes, including temperature, precipitation, snow, and impacts to water resources.
- 20 years of experience analyzing climate and hydrologic data using programming languages, statistics packages, and data visualization, including the use of R, Python, and GIS.
- 16 years of experience communicating drought and climate information to stakeholders, decision-makers, federal, state, and local partners, teachers and students, and the public.
- 18 years of experience collaborating with state climatologists, Regional Climate Centers, NOAA RISAs, and federal and academic institutions across the country.

- 18 years of experience managing websites and delivering climate data and tools.
- Led the development and publication of the Climate Change in Colorado report, in partnership with Denver Water and the Colorado Water Conservation Board.
- Contributed to state and local reports focused on assessing vulnerability, impacts, and adaptation to climate change and hazards, primarily focused on western water resources.
- Skilled at data visualization, statistical analysis and programming using Python, R, QGIS, and JavaScript.
- Experience with large hydrologic and climatic datasets includes
 - Using NOAA's North American Multi-Model Ensemble (NMME) for seasonal climate prediction with impacts to water resources.
 - Using downscaled CMIP3, CMIP5, and CMIP6 climate projections to estimate future climate impacts.
 - Modeling current and future water supply and water demand scenarios using downscaled hydrological models.
 - Analyzing current climate, drought, and water conditions using datasets from NOAA, NASA, USGS, USDA, academic centers, and state agencies.

EXTERNAL FUNDING

- Principal Investigator, August 1, 2024 – June 30, 2025: Climate Change Vulnerability Study, Phase 3; Sponsor: Colorado Water Conservation Board; Amount: \$125,000.
- Principal Investigator, January 2024 – June 30, 2028: Spur Climate Hub; Sponsor: CSU Spur; Amount: \$700,000.
- Principal Investigator, November 1, 2023 – June 30, 2023: Climate Change Vulnerability Study, Phase 2; Sponsor: Colorado Water Conservation Board; Amount: \$75,000.
- Principal Investigator, October 1, 2022 – June 30, 2023: Climate Change in Colorado report, Phase 2; Sponsor: Colorado Water Conservation Board; Amount: \$100,000.
- Principal Investigator, November 1, 2021 – June 30, 2022: Climate Change in Colorado report, Phase 1; Sponsors: Denver Water, Colorado Water Conservation Board; Amount: \$70,000.
- Principal Investigator, July 1, 2020 – June 30, 2021: Enhancing NIDIS drought monitoring and early warning in the Intermountain West; Sponsor: National Integrated Drought Information System; Amount: \$124,904.
- Principal Investigator, January 1, 2020 – September 30, 2020: Enhancing NIDIS drought monitoring and early warning in the Intermountain West; Sponsor: National Integrated Drought Information System; Amount: \$162,326
- Principal Investigator, September 1, 2019 – August 31, 2020: Identifying Drought-related Triggers and Impacts on Decision Calendars for the Ski Industry; Sponsor: NOAA, Sectoral Applications Research Program; Amount: \$70,000.

- Principal Investigator, July 1, 2018 – June 30, 2019: Enhancing NIDIS drought monitoring and early warning in the Intermountain West; Sponsor: National Integrated Drought Information System; Amount: \$220,533.00.

RECENT SPEAKING INVITATIONS

- Women in Agriculture, March 2026, Durango, CO.
- Livestock and Forage Grower Update, March 2025, Virtual.
- Rio Grande Basin Roundtable, August 2024, Alamosa, CO.
- Metro Basin Roundtable, July 2024, Denver, CO.
- Colorado Ski Country Annual Meeting, June 2024, Denver, CO.
- Yampa Basin Rendezvous, May 2024, Steamboat Springs, CO.
- Granby State of the River, May 2024, Granby, CO.
- Hydrology Seminar Series, University of Colorado, April 2024, Boulder, CO.
- Central Region Climate and Drought Update, March 2024, Virtual.
- Colorado Fruit and Veggie Grower's Association Conference, February 2024, Broomfield, CO.
- Livestock and Forage Grower Update, February 2024, Virtual.
- Southern Ag Conference, February 2024, Monte Vista, CO.
- Colorado Farm Show, January 2024, Greeley, CO.
- Colorado Crops Clinic, December 2023, Virtual.
- South Platte Forum, November 2023, Greeley, CO.
- Conservation Colorado, November 2023, Virtual.
- Colorado Renewable Energy Society, October 2023, Virtual.
- Climate Resilient Economies Roundtable, October 2023, Fort Collins, CO.
- Water Education's Water Fluency Panel Discussion, August 2023, Virtual.
- Denver Chapter of the Citizen's Climate Lobby, August 2023, Virtual.
- San Miguel Watershed Forum, July 2023, Norwood, CO.

OUTREACH

- Poudre School District Climate Leadership Summit, Fort Collins, CO, April 2023.
- Guest Lecture, Colorado State University, Sheep Systems, March 2023.
- Panelist water expert, Stem Lab School, 7th grade water sustainability projects, October 2022.
- Climate expert, Preston Middle School, Summer STEM X program, Telling Science Fact from Fiction, June 2021.
- Climate expert for outdoor industry stakeholder interviews, University of Colorado-Boulder, Masters of the Environment candidates, May 2019.
- Woman Scientist panel expert, Expanding Your Horizons Parents and Educators conference, April 2018.

- Climate change panel expert, Metropolitan State University of Denver, Hot Topic: Climate Change, March 2018.
- Actual Scientist, Senator Bernie Sanders, Climate Scientists Answer Trump Nominee's Questions, February 2018.
- Climate expert, Overland High School, Be the Change projects, November 2017.
- Climate change expert, Skyview Middle School, 8th grade letters to a scientist, March 2017.
- Weather expert, Cherrelyn Elementary, talking weather with 5th grade students, January 2017.

PUBLICATIONS

R.S. Schumacher, **R.A. Bolinger**, and J.J. Lukas, 2024: Development of Alternate Climate Divisions for Colorado Based on Gridded Data, *Journal of Applied and Service Climatology*.

R.A. Bolinger, J.J. Lukas, R.S. Schumacher, and P.E. Goble, 2024: Climate Change in Colorado, 3rd Edition. Colorado State University, <https://climatechange.colostate.edu>.

R.A. Bolinger, and Coauthors, 2022: An assessment of the extremes and impacts of the February 2021 South-Central U.S. Arctic outbreak, and how climate services can help. *Weather and Climate Extremes*.

P.E. Goble, **R.A. Bolinger**, and R.S. Schumacher, 2021: A CONUS-wide standardized precipitation-evapotranspiration index for major U.S. row crops. *Journal of Hydrometeorology*.

R. A. Bolinger, A. D. Gronewold, K. Kompoltowicz, and L. M. Fry, 2017: Application of the NMME in the Development of a New Regional Seasonal Climate Forecast Tool, *Bulletin of the American Meteorological Society*.

R. A. Bolinger, C. D. Kummerow, and N.J. Doesken, 2014: Attribution and Characteristics of Wet and Dry Seasons in the Upper Colorado River Basin, *Journal of Climate*

R. A. Smith, and C. D. Kummerow, 2013: A Comparison of in Situ, Reanalysis, and Satellite Water Budgets over the Upper Colorado River Basin, *Journal of Hydrometeorology*

K. E. Kunkel, D. R. Easterling, D. A. R. Kristovich, B. Gleason, L. Stoecker, and **R. Smith**, 2012: Meteorological Causes of the Secular Variations in Observed Extreme Precipitation Events for the Conterminous United States, *Journal of Hydrometeorology*

K. E. Kunkel, D. R. Easterling, D. A. R. Kristovich, B. Gleason, L. Stoecker, and **R. Smith**, 2010: Recent Increases in U.S. Heavy Precipitation Associated with Tropical Cyclones, *Geophysical Research Letters*

OTHER WRITING

Great Plains could see its most significant drought in a decade

Washington Post, Capital Weather Gang, published April 11, 2022

Celebrate Women's History Month with six inspiring women in atmospheric sciences

Washington Post, Capital Weather Gang, co-author, published March 8, 2022

U.S. has experienced fewer cold winter days than normal so far, a sign of climate warming

Washington Post, Capital Weather Gang, published February 23, 2022

Drought conditions improve in U.S. West, but more snow is needed

Washington Post, Capital Weather Gang, published January 22, 2022

La Niña, climate change, and bad luck: the climate context of Colorado's Marshall Fire

NOAA Climate.gov, co-author, published January 7, 2022

How extreme climate conditions fueled unprecedented Colorado fire

Washington Post, Capital Weather Gang, co-author, published December 31, 2021

Snowpack is off to a poor start in the West, bad news amid widespread drought

Washington Post, Capital Weather Gang, published December 7, 2021

Depleted by drought, Lakes Powell and Mead were doomed from the beginning

Washington Post, Capital Weather Gang, published September 10, 2021

Warming is clearly visible in new US 'climate normal' datasets

The Conversation, co-author, published May 7, 2021

Colorado River Basin's snowpack season earns low grades: Bad news for water in the West

Washington Post, Capital Weather Gang, published April 16, 2021

Did the March 2021 snowstorm improve drought conditions across Colorado?

Weather5280, published March 18, 2021

On International Women's Day, these atmospheric scientists inspire us

Washington Post, Capital Weather Gang, co-author, published March 8, 2021

Historic drought deepens in the West as window for rain, snow closes

Washington Post, Capital Weather Gang, published March 3, 2021

Last week was quite cold across much of the U.S., but was it extreme?
Weather5280, published February 22, 2021

Colorado's top weather and climate events of 2020
Weather5280, published January 1, 2021

The relationship between drought and wildfires in Colorado
Weather5280, published August 25, 2020

Derechos: what are they and can we forecast them?
Livestock Wx, published August 21, 2020

What do we mean when we talk about drought? Is it data, impacts, or both?
Livestock Wx, published July 24, 2020

Have you heard the term "flash drought?" What exactly does that mean?
Weather5280, published June 24, 2020

Reducing the current drought: historically what are the odds drought will improve this May-June?
Livestock Wx, published May 26, 2020

Drought expands across southern Colorado
Weather5280, published May 5, 2020

Let's talk about the weather, specifically nice weather!
Weather5280, published April 1, 2020

How February snows have helped shape Colorado's 2019-2020 snow season
Weather5280, published March 6, 2020

How might the new NOAA climate temperature normals change for your county?
Weather5280, published January 22, 2020

A look at Christmas Day extremes
Weather5280, published December 23, 2019

How drought prone is your state? A look at the top states and counties in drought over the last two decades
Drought.gov, published August 18, 2019
Originally published for Livestock Wx

How far away was that lightning?
The Conversation, published June 7, 2018

PRESENTATIONS

American Association of State Climatologists Annual Meeting, Blue Springs, MO, June 2025: Were We Prepared? Results from a Climate Impacts Tabletop Exercise.

American Water Resources Association Colorado Annual Meeting, Golden, CO, April 2024: Climate Change in Colorado.

AMS Annual Meeting, Baltimore, MD, January 2024: Uncertainty in Precipitation and Water Projections in the Climate Change in Colorado Report.

AMS Annual Meeting, Baltimore, MD, January 2024: Documenting Colorado's Significant Weather Events with CoAgMET.

3rd Annual Southwest Drought Learning Network Meeting, Albuquerque, NM, September 2023: Southwest Drought Recovery in 2022-23, Panel Discussion.

13th Biennial U.S. Drought Monitor Forum, Boulder City, NV, April 2023: The USDM Process and Stakeholder Feedback, Panel Moderator.

AMS Annual Meeting, Denver, CO, January 2023: Introducing Climate Services Best Practices, Town Hall Moderator.

American Association of State Climatologists Annual Meeting, Durham, NH, June 2022: A climatological perspective of the February 2021 South-Central U.S. Cold Outbreak.

AMS Weather Band Symposium, January 2022: An impacts-based assessment of the 2018 Four Corners drought using CoCoRaHS Condition Monitoring reports.

Annual Meeting of the American Association of State Climatologists, June 2021: CoAgMET – Updated maps, products, and station pages.

Conference on Major Weather Events and Societal Impacts of 2018, Phoenix, AZ, January 2019: The Evolution and Impacts of the 2018 Four Corners Drought in Southwest U.S.

Asian-Pacific Economic Cooperation (APEC) Climate Symposium, Port Moresby, Papua New Guinea, August 2018: Developing a regional drought outlook product using seasonal forecast information.

12th Symposium on Societal Applications, Seattle, WA, January 2017: Developing a Drought Outlook Product for the NIDIS Intermountain West Drought Early Warning System.

2015 American Geophysical Union Fall Meeting, Poster Session, San Francisco, CA, December 2015: Using NMME in Region-Specific Operational Seasonal Climate Forecasts.

58th Annual Conference on Great Lakes Research, Burlington, VT, May 2015: Improving Climate Inputs into Operational Lake Level Forecasts.

27th Conference on Climate Variability and Change, Poster Session, Phoenix, AZ, January 2015: Quantification of Temperature and Precipitation Variability Over the Great Lakes.

2014 American Geophysical Union Fall Meeting, Poster Session, San Francisco, CA, December 2014: Comparing Measurements, Simulations, and Forecasts of Snow Water Equivalent Across the Great Lakes Basin.

37th Climate Diagnostics and Prediction Workshop, Fort Collins, CO, October 2012: Identifying Connections Between Large-Scale Climate Variability and the Local Hydroclimate of the Upper Colorado River Basin.

26th Conference on Hydrology, New Orleans, LA, January 2012: Large-Scale Climate Variability and Local Hydroclimate Relationship Used as Drivers in a Hydrologic Model to Predict the Hydrology of the Upper Colorado River Basin.

36th Climate Diagnostics and Prediction Workshop, Poster Session, Fort Worth, TX, October 2011: The Introduction of an In-depth Drought Monitoring Process in the Upper Colorado River Basin.

19th Conference on Applied Climatology, Asheville, NC, July 2011: A Comparison of In-situ, Reanalysis, and Satellite Water Data over the Upper Colorado River Basin.

35th Annual Meeting of the American Association of State Climatologists, Poster Session, Lake Tahoe, CA, July 2010: NIDIS and the Upper Colorado River Basin Pilot: Current Monitoring Efforts and Future Research.

34th Annual Meeting of the American Association of State Climatologists, Grand Rapids, MI, July 2009: The Quality Control of the Climate Database Modernization Program's Forts Data.

52nd Annual Conference on Great Lakes Research, Toledo, OH, May 2009: The Influence of the Great Lakes on Passing Cyclones During the Stable Season.

Seventh Workshop on Decadal Climate Variability, Hilton Waikoloa Village, HI, April 2007: Trends in Maximum and Minimum Temperature Deciles in Select Regions of the United States.

19th Conference on Climate Variability and Change, 87th AMS Annual Meeting, Poster Session, San Antonio, TX, January 2007: Identifying Connections Between Temperature Trends and Station Changes in the United States.

TEACHING/MENTORING

- Climatology, MTR 3330, Metro State University of Denver, Fall 2026
- Climate of Colorado, ATS 355, Colorado State University, anticipated Spring/Fall 2027
- Master's Committee, Co-Advisor, Colorado State University, 2021 – 2025

- Mentor, Colorado State University, CAMP, 2021, 2022
- Mentor, Colorado State University, REU summer student, 2019, 2021
- Mentor, University of Michigan students, CILER summer student fellow, 2014 – 2016
- Teaching Assistant, programming for atmospheric science, Spring 2012

SERVICE

- AMS Board Certified Climate Science Professional (2026)
- AMS Applied Climatology Committee Member (2025 – present)
- AMS Climate Services Committee Chair (2019 – 2024)
- AMS Board on Enterprise Decision Support Member (2019 – 2024)
- AASC Associate-at-Large on Executive Committee (2020 – 2022)
- Search Committee Chair for CoAgMET Manager, 2023
- Search Committee for State Climatologist of Colorado, 2017
- Review Panel for NASA Water and Energy Cycle for graduate student fellowships, 2017
- Journal Peer Reviewer – Nature, Frontiers in Climate Services, Journal of Contemporary Water Research and Education, International Journal of Climatology, American Water Resources Association, Journal of Great Lakes Research, Journal of Geophysical Research

AFFILIATIONS

- Member of the American Meteorological Society
- Member of the American Association of State Climatologists
- Graduate Student Representative, CSU, 2010 – 2011
- Chi Epsilon Pi FSU Meteorology Honor Society, 2006 – 2007
- Secretary and Webmaster of the Student Chapter of the AMS, MSU-Denver, 2001 – 2002

AWARDS/GRANTS

- 2014 Postdocs Applying Climate Expertise (PACE) Fellowship (<https://cpaess.ucar.edu/pace>)
- 2013 Richard Herbert Memorial Scholarship
(https://awracolorado.org/Application_Announcement)
- 2011 Climate Diagnostics and Prediction Workshop Student Travel Grant